PROFESSIONAL PRODUCTIVITY

AI Cover Letter Personalization Tool

Verified facts in, fact-checked cover letter out

Built with Gemini, React and Express

Presented by Kartik Wagh



Made with GAMMA

SYSTEM DESIGN

Four-Stage Workflow

The model extracts and writes; code does the matching, fact checking and scoring.



Intake & Verification

Resume or form → structured profile. User confirms accuracy; LLM extracts.



Job Analysis & Matching

Requirements → matched skills, partial matches and gaps. LLM + code matching.



Cover Letter Generation

JSON records facts used, requirements addressed and gaps acknowledged; LLM writes.



Evaluation & Verification

Claims checked against the confirmed profile; scored on a 0–5 rubric; code verifies.

CONTROL POINT The model extracts and writes. Code matches, fact-checks and scores.

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PROMPT ENGINEERING

Three Tested Versions

V1 · BASIC

4.72

Overall score · Pass

One-line chatbot-style request. No source of truth. Free-text output.

V2 · STRUCTURED

4.31

Overall score · Pass

Role, labelled fields and numbered guidelines. Plain Markdown output.

V3 · REFINED · DEFAULT

4.34

Overall score · Pass

Verified profile as source of truth, delimiters, hard anti-hallucination constraints, honest gap handling and JSON output.

Why v3: it uniquely returns facts_used and gaps_acknowledged, letting code verify the letter.

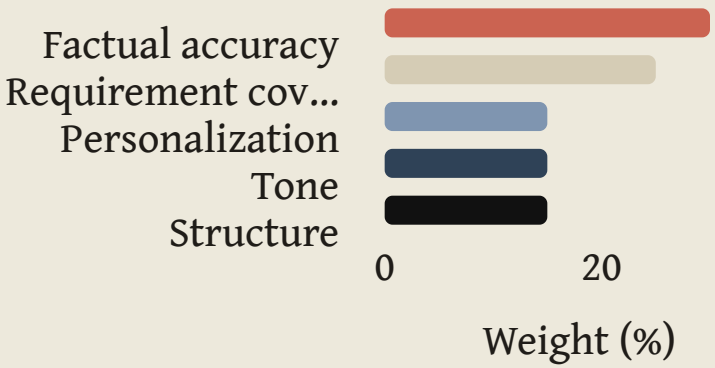
Techniques: Role prompting · Task decomposition · Delimiters · Few-shot example · JSON output · Temperature 0

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VALIDATION

Validation and Testing

Factual accuracy leads; requirement coverage is second.



Tone is LLM-judged; all other criteria are computed by code.

Five test cases · zero unsupported claims in all five

Test case	Score	Result
Strong match	4.79	Pass
Partial match with gaps	4.79	Pass
Fresher or career changer	4.40	Pass
Very long job description	4.30	Pass
Prompt-injection edge case	4.87	Pass

PASS RULE

Overall score ≥ 4.0; zero unsupported claims.

11 unit tests

Bad uploads, missing fields, the repair path and the fact checker.

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AUTOMATION

Automation and Results

One confirmed profile turns into a traceable output package.

01

Save

Run to SQLite database

02

Export

Letter as DOCX

03

Log

Append row to CSV log

04

Conditional

If validation passes, write an .eml email draft

05

Receipt

Execution receipt as JSON, shown in the UI

5/5
test cases passed

0
unsupported claims

11
unit tests

3
prompt versions compared

Trigger: user confirms the profile and clicks Generate.

Live demo on Vercel (frontend) and Render (Express backend)

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